DAVID CROPPER

AI Engineer | Data Engineer | Automation Engineer

Melbourne, Australia

Experienced engineer with a background spanning data engineering, automation engineering, cloud platforms, and operational systems. Currently focused on building practical AI-powered solutions, Retrieval-Augmented Generation (RAG) systems, workflow intelligence platforms, and governance-aware operational AI applications.

Combines enterprise experience in data pipelines, automation frameworks, cloud technologies, and operational analytics with hands-on AI engineering work covering LLMs, vector databases, retrieval systems, orchestration, evaluation, and explainable AI workflows.

Passionate about building AI systems that improve operational decision-making, automate complex workflows, and deliver trustworthy, evidence-based outputs.

Core Competencies

AI Engineering

- Retrieval-Augmented Generation (RAG)
- Large Language Models (LLMs)
- Prompt Engineering
- Query Rewriting
- Grounded AI Responses
- AI Evaluation & Telemetry
- AI Safety & Guardrails
- Confidence & Explainability Concepts
- Governance-Aware AI Design
- Operational AI Systems
- AI-Assisted Workflow Automation

Data Engineering

- Data Pipeline Development

- ETL / ELT Design
- Data Validation & Quality
- Analytics Engineering
- Operational Reporting
- Data Modelling Concepts
- Structured & Unstructured Data Processing

Programming & Development

- Python
- SQL
- PySpark
- FastAPI
- REST APIs
- JSON
- YAML
- Git / GitHub

Cloud & Platforms

- AWS
- Azure
- Databricks
- Docker
- Streamlit
- ChromaDB
- Render

Automation & Engineering

- Test Automation
- Selenium
- CI/CD Concepts
- Workflow Automation
- Systems Integration
- Operational Tooling
- Reliability & Validation Frameworks

Featured Project: Payroll Diagnostics & Governance Engine

Designed and developed a modular payroll diagnostics platform focused on identifying payroll risks, data quality issues, compliance exposures, and operational anomalies through automated analysis and governance-oriented reporting.

The platform analyses payroll datasets using configurable diagnostic rules and validation logic, generating structured findings, evidence-backed insights, and executive-level reporting to support risk visibility and operational assurance.

Key Capabilities

- Payroll data ingestion and validation
- Rule-based diagnostics and anomaly detection
- Leave liability and entitlement analysis
- Long Service Leave (LSL) reviews
- Termination compliance checks
- Record-keeping and evidence gap analysis
- Cross-module integrity validation
- Severity scoring and findings prioritisation
- Executive reporting and governance dashboards
- Coverage and data dependency analysis

Engineering Focus Areas

- Data ingestion pipelines
- Validation frameworks
- Rule engine architecture
- Operational analytics
- Evidence-backed findings generation
- Governance-aware reporting
- Automated report generation
- Modular system design
- Operational risk visibility
- Explainable and traceable outputs

Technologies

Python • PostgreSQL • Pandas • YAML • Markdown/HTML Reporting • PDF Generation • Git/GitHub

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Featured Project: Governance-Aware RAG Platform

Designed and developed a modular Retrieval-Augmented Generation (RAG) platform focused on trustworthy operational knowledge retrieval and explainable AI interactions.

Key Features

- Multi-format document ingestion (TXT, PDF, DOCX)
- Configurable chunking strategies
- Embedding generation and vector indexing
- Persistent ChromaDB vector storage
- Query rewriting and retrieval optimisation
- Source attribution and traceability
- Streamlit management interface
- Retrieval telemetry and diagnostics
- Grounded response generation
- Governance-aware AI patterns

Technologies

Python • ChromaDB • OpenAI APIs • Streamlit • Docker • FastAPI

Professional Experience

Founder

Chase Risk & Compliance (CRC)

2025 – Present

Founded a technology-focused consultancy developing governance-aware operational intelligence solutions.

Designed and built a Payroll Diagnostics & Governance Engine and a Governance-Aware RAG Platform, applying principles of operational visibility, explainability, traceability, and AI-assisted decision support.

Current focus includes Retrieval-Augmented Generation (RAG), operational AI systems, workflow intelligence, and governance-aware AI tooling.

Data Engineer

NBN Australia

2020 – 2023

Designed and developed scalable data pipelines supporting operational reporting, analytics, and business decision-making.

Worked extensively with Python, SQL, APIs, Azure technologies, Databricks, and enterprise data platforms to deliver reliable, production-grade data solutions.

Senior Test Automation Engineer / QA Automation Engineer

Designed and implemented enterprise automation frameworks and testing solutions using Selenium and modern automation practices.

Built regression, API, integration, and end-to-end automation frameworks while working closely with engineering teams to improve quality, reliability, and delivery outcomes.

Certifications

- AWS Certified Developer – Associate
- Databricks Data Engineer Associate
- Databricks Data Engineer Professional
- ISTQB Certified Tester
- Certified Agile Tester

Current Areas of Interest

- AI Engineering
- Retrieval-Augmented Generation (RAG)
- Operational AI
- AI Agents & Orchestration
- Workflow Intelligence
- Governance-Aware AI
- AI Evaluation & Reliability
- Cloud Data Engineering
- Automation Engineering
- Explainable AI Systems